FUNDAMENTALS OF DATA SCIENCE LAB

EXPERIMENT 20

K-means Clustering Algorithm

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Register Number: 250701050

Exp No:**20** Experiment to understand K-means clustering algorithm for a given data set.

Description: Understand the K-means clustering algorithm for the dataset given.

In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

In [2]:
```
df=pd.read_csv('Mall_Customers.csv')
```

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 5 columns):
 #   Column                  Non-Null Count  Dtype 
---  ------                  --------------  ----- 
 0   CustomerID              200 non-null    int64 
 1   Gender                  200 non-null    object
 2   Age                     200 non-null    int64 
 3   Annual Income (k$)      200 non-null    int64 
 4   Spending Score (1-100)  200 non-null    int64 
dtypes: int64(4), object(1)
memory usage: 7.9+ KB
```

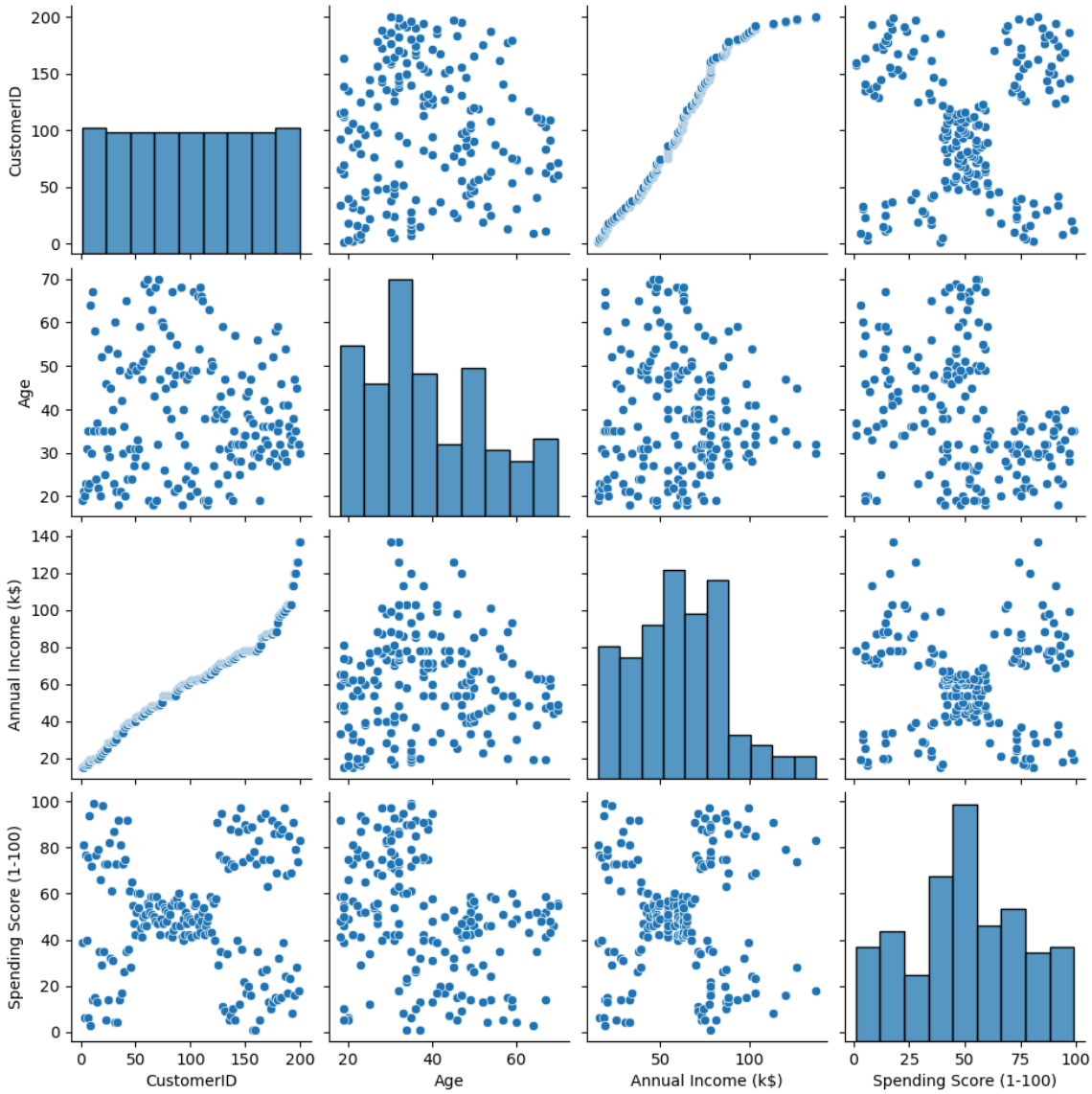
In [4]:
```
df.head()
```

Out[4]:

	CustomerID	Gender	Age	Annual Income (k$)	Spending Score (1-100)
0	1	Male	19	15	39
1	2	Male	21	15	81
2	3	Female	20	16	6
3	4	Female	23	16	77
4	5	Female	31	17	40

In [5]:
```
sns.pairplot(df)
```

Out[5]:
```
<seaborn.axisgrid.PairGrid at 0x170e8e47850>
```



In [6]:
```
features=df.iloc[:,[3,4]].values
```

In [7]:
```
from sklearn.cluster import KMeans
model=KMeans(n_clusters=5)
model.fit(features)
KMeans(n_clusters=5)
```

```
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
er\_kmeans.py:870: FutureWarning: The default value of `n_init` will chang
e from 10 to 'auto' in 1.4. Set the value of `n_init` explicitly to suppre
ss the warning
  warnings.warn(
C:\Users\Ayyadurai\AppData\Local\anaconda3\Lib\site-packages\sklearn\clust
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Windows with MKL, when there are less chunks than available threads. You c
an avoid it by setting the environment variable OMP_NUM_THREADS=1.
  warnings.warn(
```

Out[7]:
```
KMeans(n_clusters=5)
```
In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [8]:
```
Final=df.iloc[:,[3,4]]
Final['label']=model.predict(features)
Final.head()
```

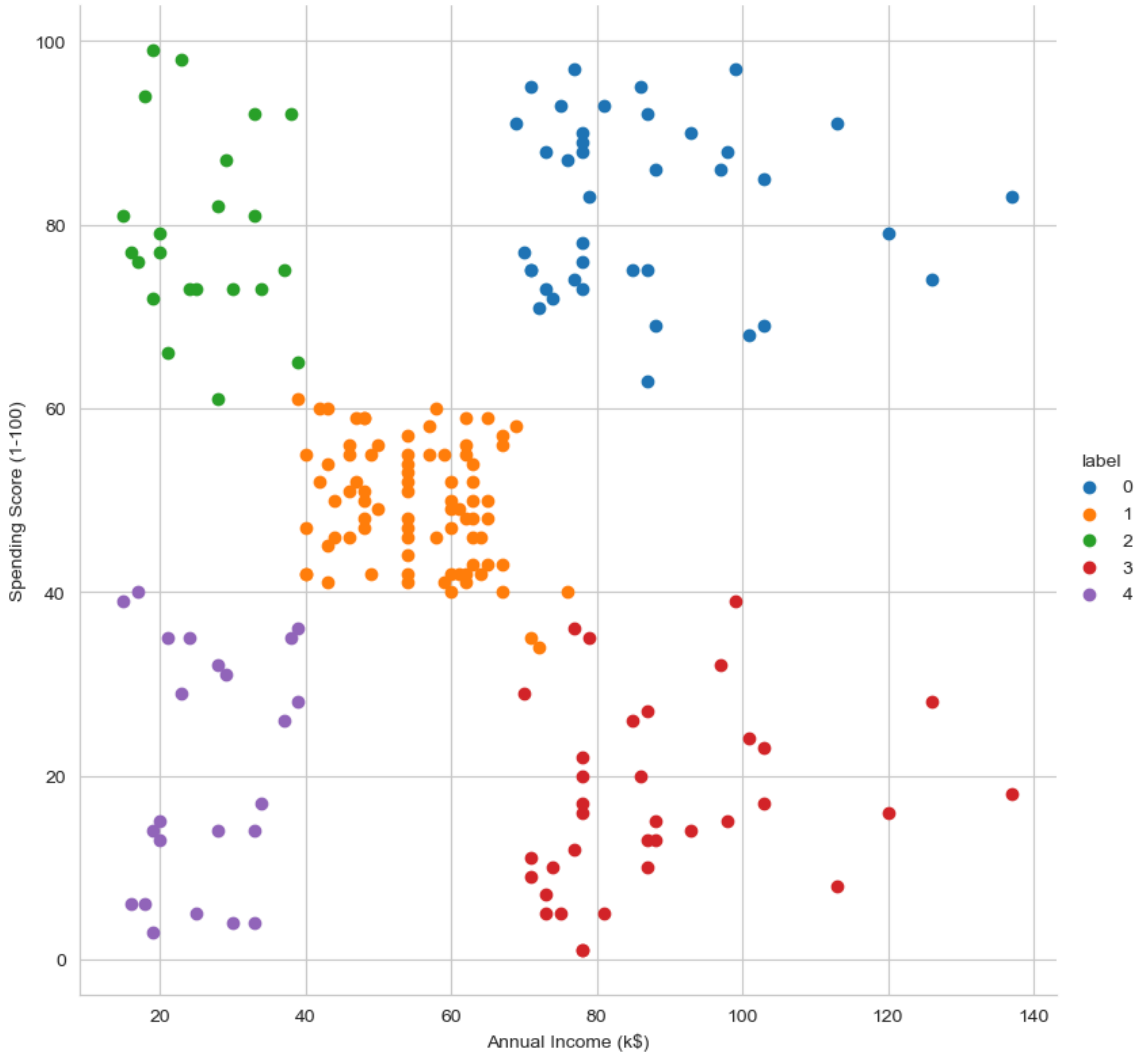
```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_8116\470183701.py:2: Setti
ngWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame.
Try using .loc[row_indexer,col_indexer] = value instead

See the caveats in the documentation: https://pandas.pydata.org/pandas-doc
s/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://
pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-
view-versus-a-copy)
  Final['label']=model.predict(features)
```

Out[8]:

	Annual Income (k$)	Spending Score (1-100)	label
0	15	39	4
1	15	81	2
2	16	6	4
3	16	77	2
4	17	40	4

In [9]:
```
sns.set_style("whitegrid")
sns.FacetGrid(Final,hue="label",height=8) \
.map(plt.scatter,"Annual Income (k$)", "Spending Score (1-100)") \
.add_legend();
plt.show()
```

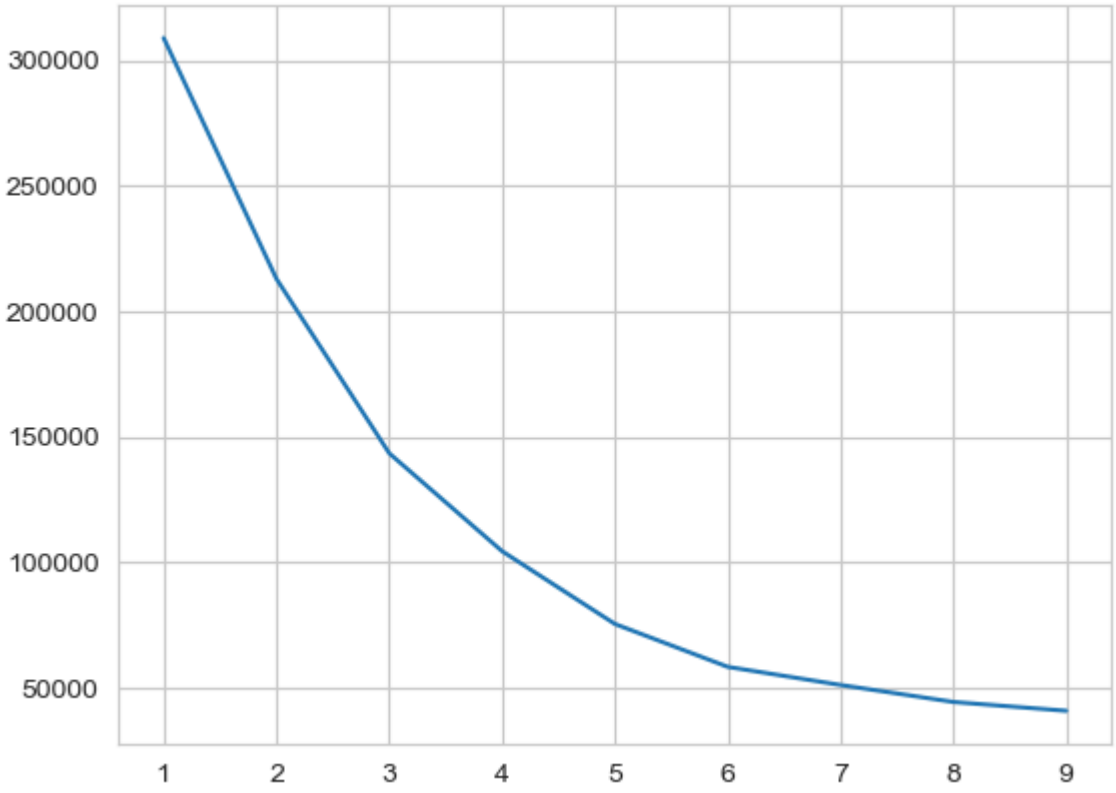


In [10]:
```
features_el=df.iloc[:,[2,3,4]].values
from sklearn.cluster import KMeans
wcss=[]
for i in range(1,10):
  model=KMeans(n_clusters=i)
  model.fit(features_el)
  wcss.append(model.inertia_)
plt.plot(range(1,10),wcss)
```

```
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```

Out[10]: `[<matplotlib.lines.Line2D at 0x170e99f3550>]`



In [ ]:

VIVA - NUMPY

1. What is Numpy?

Answer: NumPy is a Python library used for working with arrays. It also provides functions for performing mathematical and logical operations on arrays.

1. How do you install NumPy?

Answer: You can install NumPy using the pip package manager with the command pip install numpy.

2. How do you import NumPy in your code?

Answer: You can import NumPy using the command import numpy as np.

3. How do you create a NumPy array from a Python list?

Answer: You can create a NumPy array using np.array([1, 2, 3]).

4. What is the difference between a list and a NumPy array?

Answer: A list is a built-in Python data structure that can hold heterogeneous items. A NumPy array is a grid of values, all of the same type, indexed by a tuple of nonnegative integers.

5. How do you create an array of zeros in NumPy?

Answer: You can create an array of zeros using np.zeros((2, 3)).

6. How do you create an array of ones in NumPy?

Answer: You can create an array of ones using np.ones((2, 3)).

7. How do you create an array with a range of values in NumPy?

Answer: You can create an array with a range of values using np.arange(0, 10, 2).

8. What is the shape of an array?

Answer: The shape of an array is a tuple that represents the size of the array along each dimension. You can get the shape of an array using array.shape.

9. How do you reshape an array in NumPy?

Answer: You can reshape an array using the reshape method, e.g., array.reshape((2, 3)).

10. How do you get the data type of a NumPy array?

Answer: You can get the data type of an array using array.dtype.

11. How do you create a NumPy array with random values?

Answer: You can create an array with random values using np.random.rand(2, 3).

12. How do you perform element-wise addition of two arrays?

Answer: You can perform element-wise addition using the + operator, e.g., array1 + array2.

13. How do you perform matrix multiplication in NumPy?

Answer: You can perform matrix multiplication using np.dot(array1, array2) or the @ operator.

14. How do you find the maximum value in a NumPy array?

Answer: You can find the maximum value using array.max().

15. How do you find the minimum value in a NumPy array?

Answer: You can find the minimum value using array.min().

16. How do you calculate the mean of a NumPy array?

Answer: You can calculate the mean using array.mean().

17. How do you calculate the standard deviation of a NumPy array?

Answer: You can calculate the standard deviation using array.std().

18. How do you concatenate two arrays in NumPy?

Answer: You can concatenate two arrays using np.concatenate((array1, array2), axis=0).

19. How do you stack arrays vertically in NumPy?

Answer: You can stack arrays vertically using np.vstack((array1, array2)).

20. How do you stack arrays horizontally in NumPy?

Answer: You can stack arrays horizontally using np.hstack((array1, array2)).

21. How do you get the transpose of a NumPy array?

Answer: You can get the transpose using array.T.

22. How do you save a NumPy array to a file?

Answer: You can save an array to a file using np.save('filename.npy', array).

23. How do you load a NumPy array from a file?

Answer: You can load an array from a file using np.load('filename.npy').

24. What is broadcasting in NumPy?

Answer: Broadcasting is the ability of NumPy to perform arithmetic operations on arrays of different shapes by automatically expanding the smaller array to match the shape of the larger array.

MATPLOTLIB

1. What is Matplotlib?

Answer: Matplotlib is a comprehensive library for creating static, animated, and interactive visualizations in Python.

1. How do you install Matplotlib?

Answer: You can install Matplotlib using the pip package manager with the command `pip install matplotlib`.

2. How do you import Matplotlib in your code?

Answer: You can import Matplotlib using the command `import matplotlib.pyplot as plt`.

3. How do you create a simple line plot?

Answer:

```
import matplotlib.pyplot as plt
plt.plot([1, 2, 3, 4])
plt.show()
```

4. How do you add labels to the x and y axes in a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.xlabel('X-axis label')
plt.ylabel('Y-axis label')
plt.show()
```

5. How do you add a title to a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.title('Plot Title')
plt.show()
```

6. How do you plot multiple lines on the same plot?

Answer:

```
plt.plot([1, 2, 3, 4], label='Line 1')
plt.plot([4, 3, 2, 1], label='Line 2')
plt.legend()
```

7. How do you create a scatter plot?

Answer:

```
plt.scatter([1, 2, 3, 4], [10, 20, 25, 30])
plt.show()
```

8. How do you create a bar plot?

Answer:

```
plt.bar([1, 2, 3, 4], [10, 20, 15, 25])
plt.show()
```

9. How do you create a histogram?

Answer:

```
data = [1, 2, 2, 3, 3, 3, 4, 4, 4, 4]
plt.hist(data, bins=4)
plt.show()
```

10. How do you change the line style in a plot?

Answer:

```
plt.plot([1, 2, 3, 4], linestyle='--')
plt.show()
```

11. How do you change the line color in a plot?

Answer:

```
plt.plot([1, 2, 3, 4], color='red')
plt.show()
```

12. How do you change the line width in a plot?

Answer:

```
plt.plot([1, 2, 3, 4], linewidth=2)
plt.show()
```

13. How do you create a subplot with multiple plots?

Answer:

```
fig, (ax1, ax2) = plt.subplots(1, 2)
ax1.plot([1, 2, 3, 4])
ax2.plot([4, 3, 2, 1])
```

14. How do you add a grid to a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.grid(True)
plt.show()
```

15. How do you save a plot to a file?

16.

Answer:

```
plt.plot([1, 2, 3, 4])
plt.savefig('plot.png')
plt.show()
```

17. How do you create a pie chart?

Answer:

```
labels = 'A', 'B', 'C', 'D'
sizes = [15, 30, 45, 10]
plt.pie(sizes, labels=labels)
plt.show()
```

18. How do you change the figure size of a plot?

Answer:

```
plt.figure(figsize=(8, 6))
plt.plot([1, 2, 3, 4])
plt.show()
```

19. How do you create a box plot?

Answer:

```
data = [np.random.normal(0, std, 100) for std in range(1, 4)]
plt.boxplot(data)
plt.show()
```

20. How do you create a heatmap?

Answer:

```
data = np.random.rand(10, 10)
plt.imshow(data, cmap='hot', interpolation='nearest')
plt.show()
```

21. How do you add annotations to a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.annotate('Important Point', xy=(2, 2), xytext=(3, 3),
arrowprops=dict(facecolor='black', shrink=0.05))
plt.show()
```

22. How do you change the font size of labels in a plot?

Answer:

```
plt.plot([1, 2, 3, 4])
plt.xlabel('X-axis', fontsize=14)
plt.ylabel('Y-axis', fontsize=14)
plt.show()
```

23. How do you plot a logarithmic scale?

Answer:

```
plt.plot([1, 10, 100, 1000])
plt.yscale('log')
plt.show()
```

24. How do you create an error bar plot?

Answer:

```
x = [1, 2, 3, 4]
y = [2, 3, 4, 5]
errors = [0.1, 0.2, 0.3, 0.4]
plt.errorbar(x, y, yerr=errors, fmt='o')
plt.show()
```

25. How do you create a filled area plot?

Answer:

```
x = np.linspace(0, 10, 100)
y1 = np.sin(x)
y2 = np.sin(x) + 0.5
plt.fill_between(x, y1, y2)
plt.show()
```

Pandas

1. What is Pandas?

Answer: Pandas is an open-source data analysis and manipulation library for Python. It provides data structures and functions needed to manipulate structured data seamlessly.

2. How do you install Pandas?

Answer: You can install Pandas using the pip package manager with the command pip install pandas.

3. How do you import Pandas in your code?

Answer: You can import Pandas using the command import pandas as pd.

4. How do you create a Pandas DataFrame from a dictionary?

Answer:

```
import pandas as pd
data = {'Name': ['Alice', 'Bob'], 'Age': [25, 30]}
df = pd.DataFrame(data)
print(df)
```

5. How do you read a CSV file into a DataFrame?

Answer:

```
import pandas as pd
df = pd.read_csv('file.csv')
print(df)
```

6. How do you write a DataFrame to a CSV file?

Answer:

```
df.to_csv('output.csv', index=False)
```

7. How do you display the first few rows of a DataFrame?

Answer:

```
print(df.head())
```

8. How do you display the last few rows of a DataFrame?

Answer:

```
print(df.tail())
```

9. How do you get the number of rows and columns in a DataFrame?

Answer:

```
print(df.shape)
```

10. How do you get a summary of a DataFrame?

Answer:

```
print(df.describe())
```

11. How do you select a column from a DataFrame?

Answer:

```
print(df['Name'])
```

12. How do you select multiple columns from a DataFrame?

Answer:

```
print(df[['Name', 'Age']])
```

13. How do you select rows based on a condition?

Answer:

```
print(df[df['Age'] > 25])
```

14. How do you add a new column to a DataFrame?

Answer:

```
df['Salary'] = [50000, 60000]
print(df)
```

15. How do you delete a column from a DataFrame?

Answer:

```
df = df.drop(columns=['Salary'])
print(df)
```

16. How do you rename columns in a DataFrame?

Answer:

```
df = df.rename(columns={'Name': 'Employee Name'})
print(df)
```

17. How do you check for missing values in a DataFrame?

Answer:

```
print(df.isnull().sum())
```

18. How do you fill missing values in a DataFrame?

Answer:

```
df['Age'] = df['Age'].fillna(0)
print(df)
```

19. How do you drop rows with missing values?

Answer:

```
df = df.dropna()
print(df)
```

20. How do you sort a DataFrame by a column?

Answer:

```
df = df.sort_values(by='Age')
print(df)
```

21. How do you group data in a DataFrame?

Answer:

```
grouped = df.groupby('Age').size()
print(grouped)
```

22. How do you apply a function to a column in a DataFrame?

Answer:

```
df['Age'] = df['Age'].apply(lambda x: x + 1)
print(df)
```

23. How do you merge two DataFrames?

Answer:

```
df1 = pd.DataFrame({'ID': [1, 2], 'Name': ['Alice', 'Bob']})
df2 = pd.DataFrame({'ID': [1, 2], 'Salary': [50000, 60000]})
merged = pd.merge(df1, df2, on='ID')
print(merged)
```

24. How do you concatenate two DataFrames?

Answer:

```
df1 = pd.DataFrame({'Name': ['Alice'], 'Age': [25]})
df2 = pd.DataFrame({'Name': ['Bob'], 'Age': [30]})
concatenated = pd.concat([df1, df2])
print(concatenated)
```

25. How do you pivot a DataFrame?

Answer:

```
df = pd.DataFrame({
    'Date': ['2021-01-01', '2021-01-02', '2021-01-01', '2021-01-02'],
    'City': ['New York', 'New York', 'Los Angeles', 'Los Angeles'],
    'Temperature': [32, 30, 75, 72]
})
pivoted = df.pivot(index='Date', columns='City', values='Temperature')
print(pivoted)
```

TensorFlow

1. What is TensorFlow?

Answer: TensorFlow is an open-source machine learning framework developed by Google that is used for building and deploying machine learning models.

2. How do you install TensorFlow?

Answer: You can install TensorFlow using the pip package manager with the command pip install tensorflow.

3. How do you import TensorFlow in your code?

Answer: You can import TensorFlow using the command import tensorflow as tf.

4. How do you create a constant tensor in TensorFlow?

Answer:

```
import tensorflow as tf
tensor = tf.constant([1, 2, 3])
print(tensor)
```

5. How do you create a variable tensor in TensorFlow?

Answer:

```
tensor = tf.Variable([1, 2, 3])
print(tensor)
```

6. How do you perform element-wise addition of two tensors?

Answer:

```
a = tf.constant([1, 2, 3])
b = tf.constant([4, 5, 6])
result = tf.add(a, b)
print(result)
```

7. How do you perform matrix multiplication in TensorFlow?

Answer:

```
a = tf.constant([[1, 2], [3, 4]])
b = tf.constant([[5, 6], [7, 8]])
result = tf.matmul(a, b)
print(result)
```

8. How do you create a tensor filled with zeros?

Answer:

```
tensor = tf.zeros([2, 3])
print(tensor)
```

9. How do you create a tensor filled with ones?

Answer:

```
tensor = tf.ones([2, 3])
print(tensor)
```

10. How do you create a tensor with a specific shape and value?

Answer:

```
tensor = tf.fill([2, 3], value=9)
print(tensor)
```

11. How do you reshape a tensor?

Answer:

```
tensor = tf.constant([[1, 2, 3], [4, 5, 6]])
reshaped = tf.reshape(tensor, [3, 2])
print(reshaped)
```

12. What is the purpose of the tf.function decorator?

Answer: The tf.function decorator converts a Python function into a TensorFlow graph for faster execution.

13. How do you create a simple sequential model in TensorFlow?

Answer:

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense

model = Sequential([
    Dense(64, activation='relu', input_shape=(32,)),
    Dense(10, activation='softmax')
])
```

14. How do you compile a model in TensorFlow?

Answer:

```
model.compile(optimizer='adam',  loss='sparse_categorical_crossentropy',  metrics=['accuracy'])
```

15. How do you fit a model to training data in TensorFlow?

Answer:

```
model.fit(x_train, y_train, epochs=10, batch_size=32)
```

16. How do you evaluate a model on test data in TensorFlow?

Answer:

```
model.evaluate(x_test, y_test)
```

17. How do you make predictions with a trained model in TensorFlow?

Answer:

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```
predictions = model.predict(x_new)
```

18. What is TensorFlow Datasets (TFDS)?

Answer: TensorFlow Datasets (TFDS) is a collection of ready-to-use datasets for use with TensorFlow, making it easier to import and work with standard datasets.

19. How do you load the MNIST dataset using TensorFlow Datasets?

Answer:

```
import tensorflow_datasets as tfds
mnist = tfds.load('mnist', split='train', as_supervised=True)
```

20. What is the purpose of tf.data API?

Answer: The tf.data API provides utilities for creating efficient input pipelines, handling large datasets, and performing preprocessing steps.

21. How do you create a dataset from a tensor in TensorFlow?

Answer:

```
data = tf.constant([1, 2, 3, 4, 5])
dataset = tf.data.Dataset.from_tensor_slices(data)
```

22. How do you batch elements of a dataset in TensorFlow?

Answer:

```
batched_dataset = dataset.batch(2)
```

23. How do you shuffle elements of a dataset in TensorFlow?

Answer:

```
shuffled_dataset = dataset.shuffle(buffer_size=10)
```

24. How do you repeat a dataset in TensorFlow?

Answer:

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repeated_dataset = dataset.repeat(3)
```

25. How do you save and load a TensorFlow model?

Answer:

```
# Saving the model
model.save('my_model.h5')

# Loading the model
loaded_model = tf.keras.models.load_model('my_
```